Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	McKinley, Gareth Ph.D.		
Project Title:	Resin-Spinning of Synthetic Polymer Fibers with Silk-Like Properties		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	01/01/1999	**End Date:**	01/01/2003
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Despite significant advances in understanding of the amino-acid sequence and secondary and tertiary molecular structure of a number of silk proteins, there are no spider silks commercially available, principally due to the difficulty in spinning the material. Silk fibers naturally spun by a spider are in fact complex engineering systems with a hierarchy of internal structures and orientation optimized to provide a unique combination of high strength and flexibility. This multi-level structure is generated in vivo by the spinneret structures of the orb-weaving spiders. Analyses of natural silk by nuclear magnetic resonance (NMR) or atomic force microscopy (AFM) probes confirm this complex internal structure; however, it has been impossible to date to mimic this natural spinning process. In this research we aim on developing a fundamental understanding of the processes by which natural silks are successfully spun.
In this research we are focusing on a two-pronged approach to both synthesizing and spinning new polymeric materials into structurally graded fibers which will have similar ultimate mechanical properties to that of protein polymer silks. We propose a novel and systematic approach to spinning such high-molecular-weight concentrated polymeric fluids which is

Task Description:	'outside the box' of the standard parameter space characterizing the 'dry-spinning' or wet-spinning' techniques conventionally employed in industry. We refer to this technology as 'resin-spinning' and has its scientific origin in a fundamental understanding of non-Newtonian fluid mechanics. Specifically, by controlling the rheological properties of the polymer in solution it is possible to enhance the key mechanisms of shear-induced orientation in the spinneret and also the large additional molecular orientation induced during spinning. This permits spinning of fibers with a graded microstructural orientation that varies radially through the rod in the same manner as observed in natural spider. The key is to throughly understand the role of processing history in the production of silk-like fibers. Spider drag-line silk is an engineering material which exhibits a high tensile strength plus a remarkable toughness which may exceed that of steel by 100-fold. Its flexibility, elasticity and resistance to break thus make it an ideal material for a number of the specific applications highlighted in this NASA Research Announcement (NRA) (e.g., inflatable bladders with high resistance to tearing and puncturing).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period--end date is 1/1/2003 so project is over.
Bibliography Type:	Description: (Last Updated: 09/04/2025)